

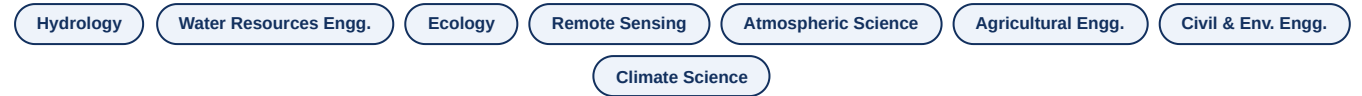
Data Science, Statistics & ML · Types of Data · Concluding Remarks

Concise exam-oriented revision sheet — concepts strictly as covered in lecture

1 Statistical & Machine Learning in Interdisciplinary Studies



Data science sits between raw scientific data and the methods used to analyze it — **statistics** and **machine learning** are the core tool-sets data science draws on to extract insight from scientific data. Data science approaches are applied across many scientific/engineering domains to analyze domain-specific data.



NOTE Different scientific/engineering domains have **different data characteristics** — data type, scale, and behaviour vary by domain, so data science methods must be adapted rather than applied identically everywhere.

2 Types of Data

DISCRETE DATA Distinct, countable values (typically integers); no meaningful values between two consecutive counts. - Number of rainy days in a month - Digital number of an image - Inter-arrival time in days	CATEGORICAL DATA Represents distinct groups or labels rather than numeric measurements — a category, not a quantity. - Flow category: low, medium, high - Rainy / non-rainy days - Drought categories - Extreme / non-extreme weather conditions	CONTINUOUS DATA Can take any value within a range, including fractions — measured rather than counted. - Daily temperature, air pressure - Wind speed, rainfall depth - Streamflow magnitude - Crop yield, sea-level rise
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Type	Nature of Values	How It Differs
Discrete	Distinct, countable numeric values	Countable in whole units; no meaningful in-between values
Categorical	Group / class labels	Represents a category, not a measured quantity
Continuous	Any value in a range, incl. fractions	Measured on a continuous scale; can take infinitely many values

3 Concluding Remarks

INTERDISCIPLINARY DATA SCIENCE VS TRADITIONAL ML Interdisciplinary scientific data combines discrete , categorical , and continuous variables with domain-specific behaviour. Since data characteristics differ across domains, generic ML approaches may not directly transfer — methods must be adapted to the scientific data at hand.	EXTRAPOLATION RISK When a model meets unprecedented conditions or extreme outliers beyond its training range, it must extrapolate rather than interpolate — with no direct evidence of behaviour there, such predictions are less reliable.
CONTINUOUS / UNBOUNDED VARIABLES Wind speed and temperature are continuous and effectively unbounded — they can take a wide range of values, including values not previously observed in training data.	NONLINEAR ML: SHORT-TERM VS LONG-TERM Nonlinear ML models perform well for short-term forecasting , where nonlinear relationships stay intact. Over longer horizons, statistical averaging over time smooths/weakens these relationships, reducing model effectiveness for long-term data.

Key Takeaway

Interdisciplinary data science must handle mixed data types, unbounded continuous variables, extrapolation risk at extremes, and nonlinear relationships that weaken over long horizons due to statistical averaging.

QUICK REVISION	
1. Data science links scientific data to statistics & machine learning for insight extraction.	6. Continuous : measured values on a range (temperature, pressure, wind speed, rainfall, streamflow, crop yield, sea-level rise).
2. Applied across Hydrology, Water Resources, Ecology, Remote Sensing, Atmospheric Science, Agricultural Engg., Civil & Env. Engg., Climate Science.	7. Unbounded continuous variables (wind speed, temperature) risk extrapolation at extremes/unprecedented conditions.
3. Different domains have different data characteristics.	8. Nonlinear ML models: strong short-term, weaker long-term as statistical averaging smooths nonlinear relationships.
4. Discrete : countable values (rainy days/month, digital number, inter-arrival time).	
5. Categorical : group labels (flow category, rainy/non-rainy, drought category, extreme/non-extreme).	